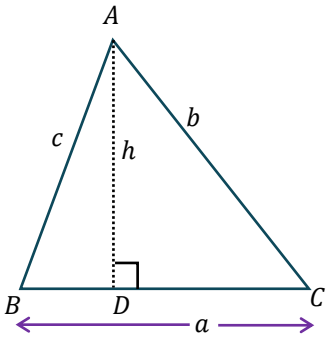
03 Solution of Triangles

The process of calculating the sides and angles of triangle using given information is called solution of triangle.

In a ΔABC, the angles are denoted by capital letters A, B and C and the length of the sides opposite these angle are denoted by small letter a, b and c respectively.

1. SINE FORMULAE:



In any triangle ABC

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=\lambda=\frac{abc}{2\Delta}=2R$$

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where R is circumradius and Δ is area of triangle.

Illustration 1

Angles of a triangle are in 4 : 1 : 1 ratio. The ratio between its greatest side and perimeter is

(A) $\frac{3}{2+\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2-\sqrt{3}}$ (D) $\frac{1}{2+\sqrt{3}}$

Ans. (B)

Solution:

Angles are in ratio 4 : 1 : 1.

$\Rightarrow$ angles are 120°, 30°, 30°.

If sides opposite to these angles are a, b, c respectively, then a will be the greatest side. Now from sine formula $\frac{a}{\sin 120°}=\frac{b}{\sin 30°}=\frac{c}{\sin 30°}$

$\Rightarrow \frac{a}{\sqrt{3}/2}=\frac{b}{1/2}=\frac{c}{1/2}$

$\Rightarrow \frac{a}{\sqrt{3}}=\frac{b}{1}=\frac{c}{1}=k$ (say)

then $a=\sqrt{3}k$, perimeter = $(2+\sqrt{3})k$

$\therefore$ required ratio = $\frac{\sqrt{3}k}{(2+\sqrt{3})k}=\frac{\sqrt{3}}{2+\sqrt{3}}$

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Illustration 2

In triangle ABC, if $b = 3, c = 4$ and $\angle B = \pi/3$, then number of such triangles is -

(A) 1 (B) 2 (C) 0 (D) infinite

Ans. (C)

Solution:

Using sine formulae $\frac{\sin B}{b} = \frac{\sin C}{c}$

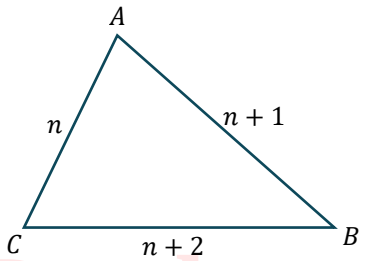
$\Rightarrow \frac{\sin \pi/3}{3} = \frac{\sin C}{4} \Rightarrow \frac{\sqrt{3}}{6} = \frac{\sin C}{4} \Rightarrow \sin C = \frac{2}{\sqrt{3}} > 1$ which is not possible.

Hence there exist no triangle with given elements.

Illustration 3

The sides of a triangle are three consecutive natural numbers and its largest angle is twice the smallest one. Determine the sides of the triangle.

Solution:



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Let the sides be $n, n+1, n+2$ *cms*.

i.e. $AC = n, AB = n+1, BC = n+2$

Smallest angle is B and largest one is A.

Here, $\angle A = 2\angle B$

Also, $\angle A + \angle B + \angle C = 180°$

$\Rightarrow 3\angle B + \angle C = 180° \Rightarrow \angle C = 180° - 3\angle B$

We have, sine law as,

$$\frac{\sin A}{n+2} = \frac{\sin B}{n} = \frac{\sin C}{n+1} \Rightarrow \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin(180-3B)}{n+1}$$

$$\Rightarrow \underset{(i)}{\frac{\sin 2B}{n+2}} = \underset{(ii)}{\frac{\sin B}{n}} = \underset{(iii)}{\frac{\sin 3B}{n+1}}$$

from (i) and (ii);

$$\frac{2\sin B\cos B}{n+2} = \frac{\sin B}{n} \Rightarrow \cos B = \frac{n+2}{2n} \quad \text{...(iv)}$$

and from (ii) and (iii);

$$\frac{\sin B}{n} = \frac{3\sin B - 4\sin^3 B}{n+1} \Rightarrow \frac{\sin B}{n} = \frac{\sin B(3-4\sin^2 B)}{n+1}$$

$$\Rightarrow \frac{n+1}{n} = 3 - 4(1-\cos^2 B) \quad \text{...(v)}$$

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from (iv) and (v), we get

$\frac{n+1}{n}=-1+4\left(\frac{n+2}{2n}\right)^2 \Rightarrow \frac{n+1}{n}+1=\left(\frac{n^2+4n+4}{n^2}\right)$

$\Rightarrow \frac{2n+1}{n}=\frac{n^2+4n+4}{n^2} \Rightarrow 2n^2+n=n^2+4n+4$

$\Rightarrow n^2-3n-4=0 \Rightarrow (n-4)(n+1)=0$

$n = 4$ or -1

where $n \neq -1$

$\therefore n = 4$. Hence the sides are 4, 5, 6

2. COSINE FORMULAE:

(a) $\cos A=\frac{b^2+c^2-a^2}{2bc}$ or $a^2 = b^2 + c^2 - 2bc\cos A$

(b) $\cos B=\frac{c^2+a^2-b^2}{2ca}$

(c) $\cos C=\frac{a^2+b^2-c^2}{2ab}$

Illustration 4

In a triangle ABC, if $B = 30°$ and $c = \sqrt{3}\,b$, then A can be equal to -

(A) 45° (B) 60° (C) 90° (D) 120°

Ans. (C)

Solution:

We have $\cos B=\frac{c^2+a^2-b^2}{2ca}$

$\Rightarrow \frac{\sqrt{3}}{2}=\frac{3b^2+a^2-b^2}{2\times\sqrt{3}b\times a}$

$\Rightarrow a^2 - 3ab + 2b^2 = 0$

$\Rightarrow (a - 2b)\,(a - b) = 0$

$\Rightarrow$ Either $a = b \Rightarrow A = 30°$

or $a = 2b \Rightarrow a^2 = 4b^2 = b^2 + c^2 \Rightarrow A = 90°$.

Illustration 5

In a triangle ABC, $(a^2 - b^2 - c^2)\tan A + (a^2 - b^2 + c^2)\tan B$ is equal to -

(A) $(a^2 + b^2 - c^2)\tan C$ (B) $(a^2 + b^2 + c^2)\tan C$

(C) $(b^2 + c^2 - a^2)\tan C$ (D) none of these

Ans. (D)

Solution:

Using cosine law :

The given expression is equal to $-2\,bc\cos A\tan A + 2\,ac\cos B\tan B$

$= 2abc\left(-\frac{\sin A}{a}+\frac{\sin B}{b}\right)=0$

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Illustration 6

In a triangle ABC, A, B, C are in A.P. Show that $2\cos\left(\frac{A-C}{2}\right)=\frac{a+c}{\sqrt{a^2-ac+c^2}}$.

Solution:

$A + C = 2B \Rightarrow A + B + C = 3B \Rightarrow B = 60°$

$\therefore \cos 60° = \frac{a^2+c^2-b^2}{2ac} \Rightarrow a^2 - ac + c^2 = b^2$

$$\Rightarrow \frac{a+c}{\sqrt{a^2-ac+c^2}} = \frac{a+c}{b} = \left[\frac{\sin A+\sin C}{\sin B}\right] = \frac{2\sin\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right)}{\sin B}$$

$= 2\cos\frac{A-C}{2}$ $\quad (\because A + C = 2B)$

Illustration 7

If in a triangle ABC, $\frac{\cos A + 2\cos C}{\cos A + 2\cos B} = \frac{\sin B}{\sin C}$ prove that the triangle ABC is either isosceles or right angled.

Solution:

$\cos A(\sin B - \sin C) + (\sin 2B - \sin 2C) = 0$

$\Rightarrow \cos A.(\sin B - \sin C) + 2\cos(B + C)\sin(B - C) = 0$ $\quad \because B + C = \pi - A$

$\Rightarrow \cos A.(\sin B - \sin C) - 2\cos A.\sin(B - C) = 0$

$\Rightarrow \cos A[(\sin B - \sin C) - 2(\sin B\cos C - \cos B\sin C)] = 0$

$\Rightarrow$ either $\cos A = 0 \Rightarrow A = 90° \Rightarrow$ right angled

or $(\sin B - \sin C) - 2(\sin B\cos C - \cos B\sin C) = 0$

$$\Rightarrow (b - c) - 2\left(b.\frac{a^2+b^2-c^2}{2ab} - c.\frac{a^2+c^2-b^2}{2ac}\right) = 0$$

$\Rightarrow a(b - c) - 2(b^2 - c^2) = 0$

$(b - c)[a - 2(b + c)] = 0$

$\therefore b - c = 0 \Rightarrow b = c \quad \Rightarrow$ isosceles

3. PROJECTION FORMULAE:

(a) $b\cos C + c\cos B = a$

(b) $c\cos A + a\cos C = b$

(c) $a\cos B + b\cos A = c$

Illustration 8

In a ΔABC, $c\cos^2\frac{A}{2} + a\cos^2\frac{C}{2} = \frac{3b}{2}$, then show a, b, c are in A.P.

Solution:

Here, $\frac{c}{2}(1+\cos A)+\frac{a}{2}(1+\cos C)=\frac{3b}{2}$

$\Rightarrow a + c + (c\cos A + a\cos C) = 3b$

$\Rightarrow a + c + b = 3b$ $\quad$ {using projection formula}

$\Rightarrow a + c = 2b$

which shows a, b, c are in A.P.

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4. NAPIER'S ANALOGY (TANGENT RULE):

(a) $\tan\left(\frac{B-C}{2}\right)=\frac{b-c}{b+c}\cot\frac{A}{2}$

(b) $\tan\left(\frac{C-A}{2}\right)=\frac{c-a}{c+a}\cot\frac{B}{2}$

(c) $\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}\cot\frac{C}{2}$

Illustration 9

In a ΔABC, the tangent of half the difference of two angles is one-third the tangent of half the sum of the angles. Determine the ratio of the sides opposite to the angles.

Solution:

Here, $\tan\left(\frac{A-B}{2}\right)=\frac{1}{3}\tan\left(\frac{A+B}{2}\right)$ ...(i)

using Napier's analogy, $\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right)$ ...(ii)

from (i) & (ii) ;

$\frac{1}{3}\tan\left(\frac{A+B}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right)\Rightarrow\frac{1}{3}\cot\left(\frac{C}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right)$

{as $A+B+C=\pi$ $\quad\therefore\tan\left(\frac{B+C}{2}\right)=\tan\left(\frac{\pi}{2}-\frac{C}{2}\right)=\cot\frac{C}{2}$}

$\Rightarrow\frac{a-b}{a+b}=\frac{1}{3}$ or $3a-3b=a+b$

$2a=4b$ or $\frac{a}{b}=\frac{2}{1}\Rightarrow\frac{b}{a}=\frac{1}{2}$

Thus the ratio of the sides opposite to the angles is $b:a=1:2$. **Ans.**

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5. HALF ANGLE FORMULAE:

$s=\frac{a+b+c}{2}$ = semi-perimeter of triangle.

(a) (i) $\sin\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{bc}}$ (ii) $\sin\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{ca}}$ (iii) $\sin\frac{C}{2}=\sqrt{\frac{(s-a)(s-b)}{ab}}$

(b) (i) $\cos\frac{A}{2}=\sqrt{\frac{s(s-a)}{bc}}$ (ii) $\cos\frac{B}{2}=\sqrt{\frac{s(s-b)}{ca}}$ (iii) $\cos\frac{C}{2}=\sqrt{\frac{s(s-c)}{ab}}$

(c) (i) $\tan\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}=\frac{\Delta}{s(s-a)}$ (ii) $\tan\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{s(s-b)}}=\frac{\Delta}{s(s-b)}$ (iii) $\tan\frac{C}{2}=\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}=\frac{\Delta}{s(s-c)}$

(d) Area of Triangle

$\Delta=\sqrt{s(s-a)(s-b)(s-c)}=\frac{1}{2}bc\sin A=\frac{1}{2}ca\sin B=\frac{1}{2}ab\sin C=\frac{1}{2}ap_1=\frac{1}{2}bp_2=\frac{1}{2}cp_3$,

where p_1,p_2,p_3 are altitudes from vertices A,B,C respectively.

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Illustration 10

If in a triangle ABC, CD is the angle bisector of the angle ACB, then CD is equal to-

(A) $\frac{a+b}{2ab}\cos\frac{C}{2}$ (B) $\frac{2ab}{a+b}\sin\frac{C}{2}$ (C) $\frac{2ab}{a+b}\cos\frac{C}{2}$ (D) $\frac{b\sin\angle DAC}{\sin(B+C/2)}$

Ans. (C, D)

Solution:

$\Delta CAB = \Delta CAD + \Delta CDB$

$\Rightarrow \frac{1}{2}ab\sin C = \frac{1}{2}b.CD.\sin\left(\frac{C}{2}\right)+\frac{1}{2}a.CD\sin\left(\frac{C}{2}\right)$

$\Rightarrow CD(a+b)\sin\left(\frac{C}{2}\right) = ab\left(2\sin\left(\frac{C}{2}\right)\cos\left(\frac{C}{2}\right)\right)$

So $CD = \frac{2ab\cos(C/2)}{(a+b)}$

and in ΔCAD, $\frac{CD}{\sin\angle DAC} = \frac{b}{\sin\angle CDA}$ (by sine rule)

$\Rightarrow CD = \frac{b\sin\angle DAC}{\sin(B+C/2)}$

Illustration 11

If Δ is the area and $2s$ the sum of the sides of a triangle, then show $\Delta \le \frac{s^2}{3\sqrt{3}}$.

Solution:

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We have, $2s = a+b+c, \Delta^2 = s(s-a)(s-b)(s-c)$

Now, A.M. $\ge$ G.M.

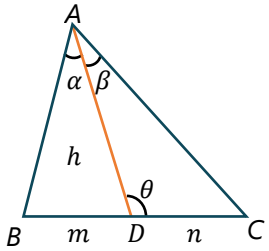
$\frac{(s-a)+(s-b)+(s-c)}{3} \ge \{(s-a)(s-b)(s-c)\}^{1/3}$

or $\frac{3s-2s}{3} \ge \left(\frac{\Delta^2}{s}\right)^{1/3}$

or $\frac{s}{3} \ge \left(\frac{\Delta^2}{s}\right)^{1/3}$

or $\frac{\Delta^2}{s} \le \frac{s^3}{27} \Rightarrow \Delta \le \frac{s^2}{3\sqrt{3}}$ **Ans**

6. $m-n$ THEOREM:



$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$

$(m+n)\cot\theta = n\cot B - m\cot C.$

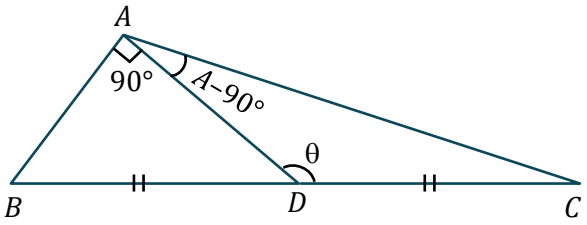
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Illustration 12

In a ΔABC . AD divides BC in the ratio 2 : 1 such that at $\angle BAD = 90°$ then prove that $\tan A + 3\tan B = 0$

Solution:

From the figure , we see that $\theta = 90° + B$ (as θ is external angle of ΔABD)



Now if we apply **m-n rule** in ΔABC, we get

$(2 + 1)\cot(90° + B) = 2.\cot 90° - 1.\cot(A - 90°)$

$\Rightarrow -3\tan B = \cot(90° - A)$

$\Rightarrow -3\tan B = \tan A$

$\Rightarrow \tan A + 3\tan B = 0$

Illustration 13

The base of a Δ is divided into three equal parts. If α, β, γ be the angles subtended by these parts at the vertex, prove that :

$(\cot\alpha + \cot\beta)(\cot\beta + \cot\gamma) = 4\operatorname{cosec}^2\beta$

Solution:

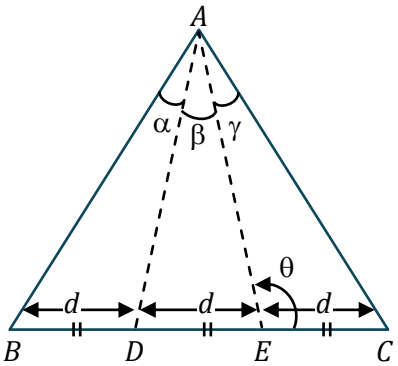
Let point D and E divides the base BC into three equal parts i.e. $BD = DE = EC = d$ (Let) and let α, β and γ be the angles subtended by BD, DE and EC respectively at their opposite vertex. Now in ΔABC

$\because BE : EC = 2d : d = 2 : 1$

$\therefore$ from ***m-n*** **rule**, we get

$(2 + 1)\cot\theta = 2\cot(\alpha + \beta) - \cot\gamma$

$\Rightarrow 3\cot\theta = 2\cot(\alpha + \beta) - \cot\gamma$ ...(i)



again

$\because$ in ΔADC

$\because DE : EC = d : d = 1 : 1$

$\therefore$ if we apply ***m-n*** **rule** in ΔADC, we get

$(1 + 1)\cot\theta = 1.\cot\beta - 1\cot\gamma$

$2\cot\theta = \cot\beta - \cot\gamma$ ...(ii)

from (i) and (ii), we get $\dfrac{3\cot\theta}{2\cot\theta} = \dfrac{2\cot(\alpha+\beta)-\cot\gamma}{\cot\beta-\cot\gamma}$

$\Rightarrow 3\cot\beta - 3\cot\gamma = 4\cot(\alpha + \beta) - 2\cot\gamma$

$\Rightarrow 3\cot\beta - \cot\gamma = 4\cot(\alpha + \beta)$

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$\Rightarrow 3\cot\beta - \cot\gamma = 4\left\{\frac{\cot\alpha.\cot\beta - 1}{\cot\beta + \cot\alpha}\right\}$

$\Rightarrow 3\cot^2\beta + 3\cot\alpha\cot\beta - \cot\beta\cot\gamma - \cot\alpha\cot\gamma = 4\cot\alpha\cot\beta - 4$

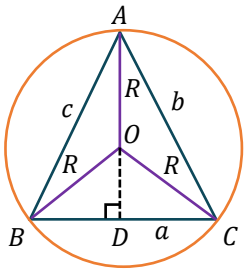
$\Rightarrow 4 + 3\cot^2\beta = \cot\alpha\cot\beta + \cot\beta\cot\gamma + \cot\alpha\cot\gamma$

$\Rightarrow 4 + 4\cot^2\beta = \cot\alpha\cot\beta + \cot\alpha\cot\gamma + \cot\beta\cot\gamma + \cot^2\beta$

$\Rightarrow 4(1 + \cot^2\beta) = (\cot\alpha + \cot\beta)(\cot\beta + \cot\gamma)$

$\Rightarrow (\cot\alpha + \cot\beta)(\cot\beta + \cot\gamma) = 4\operatorname{cosec}^2\beta$

7. RADIUS OF THE CIRCUMCIRCLE 'R':



Circumcentre is the point of intersection of perpendicular bisectors of the sides and distance between circumcentre & vertex of triangle is called circumradius 'R'.

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}.$$

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Illustration 14

In a ΔABC , prove that $\sin 2A + \sin 2B + \sin 2C = 2\Delta/R^2$

Solution:

In a ΔABC, we know that $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

and $\sin 2A + \sin 2B + \sin 2C = 4\sin A\sin B\sin C$

$= \frac{4abc}{8R^3} = \frac{16\Delta R}{8R^3} = \frac{2\Delta}{R^2}$

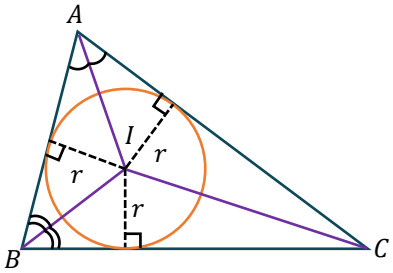
8. RADIUS OF THE INCIRCLE 'r':

Point of intersection of internal angle bisectors is incentre and perpendicular distance of incentre from any side is called inradius 'r'.

$$r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2}$$

$$= (s-c)\tan\frac{C}{2} = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$$

$$= a\frac{\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}} = b\frac{\sin\frac{A}{2}\sin\frac{C}{2}}{\cos\frac{B}{2}} = c\frac{\sin\frac{B}{2}\sin\frac{A}{2}}{\cos\frac{C}{2}}$$



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Illustration 15

In a triangle ABC, if $a:b:c = 4:5:6$, then ratio between its circumradius and inradius is-

(A) $\frac{16}{7}$ (B) $\frac{16}{9}$ (C) $\frac{7}{16}$ (D) $\frac{11}{7}$

Ans. (A)

Solution:

$$\frac{R}{r}=\frac{abc}{4\Delta}\Big/\frac{\Delta}{s}=\frac{(abc)s}{4\Delta^2} \Rightarrow \frac{R}{r}=\frac{abc}{4(s-a)(s-b)(s-c)} \quad ...(i)$$

$\because a:b:c = 4:5:6 \Rightarrow \frac{a}{4}=\frac{b}{5}=\frac{c}{6}=k$ (say)

$\Rightarrow a = 4k, b = 5k, c = 6k$

$\therefore\ s=\frac{a+b+c}{2}=\frac{15k}{2}, s-a=\frac{7k}{2}, s-b=\frac{5k}{2}, s-c=\frac{3k}{2}$

using (i) in these values $\frac{R}{r}=\frac{(4k)(5k)(6k)}{4\left(\frac{7k}{2}\right)\left(\frac{5k}{2}\right)\left(\frac{3k}{2}\right)}=\frac{16}{7}$

Illustration 16

If A, B, C are the angles of a triangle, prove that : $\cos A + \cos B + \cos C = 1+\frac{r}{R}$.

Solution:

$$\cos A + \cos B + \cos C = 2\cos\left(\frac{A+B}{2}\right).\cos\left(\frac{A-B}{2}\right)+\cos C$$

$$=2\sin\frac{C}{2}.\cos\left(\frac{A-B}{2}\right)+1-2\sin^2\frac{C}{2} = 1+2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right)-\sin\left(\frac{C}{2}\right)\right]$$

$$= 1+2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right] \qquad \left\{\because \frac{C}{2}=90°-\left(\frac{A+B}{2}\right)\right\}$$

$$= 1+2\sin\frac{C}{2}.2\sin\frac{A}{2}.\sin\frac{B}{2}=1+4\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2}$$

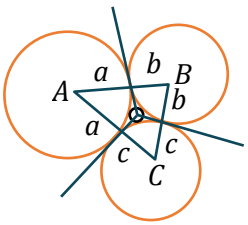
$$= 1+\frac{r}{R} \qquad \{\text{as, } r = 4R\sin A/2\,.\,\sin B/2\,.\,\sin C/2\}$$

$\Rightarrow \cos A + \cos B + \cos C = 1+\frac{r}{R}$. Hence proved.

Illustration 17

Three circles, whose radii are a, b and c, touch one another externally and the tangents at their points of contact meet in a point, prove that the distance of this point from either of their points of contact is $\left(\frac{abc}{a+b+c}\right)^{\frac{1}{2}}$.

Solution:



required distance = inradius of ΔABC

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$\because\ 2s = a+b+b+c+c+a$

$= 2(a+b+c)$

$s = a+b+c$

$\Delta = \sqrt{s(s-(a+b))(s-(b+c))\quad(s-(c+a))} = \sqrt{(a+b+c)(abc)}$

$\therefore$ required distance

$= \frac{\Delta}{s} = \frac{\sqrt{(a+b+c)(abc)}}{(a+b+c)} = \sqrt{\frac{abc}{a+b+c}} = \left(\frac{abc}{a+b+c}\right)^{\frac{1}{2}}$

9. RADIUS OF THE EX- CIRCLES:

If r_1, r_2, r_3 are the radii of the ex-circles of ΔABC opposite to the vertex A, B, C respectively, then

(i) $r_1 = \frac{\Delta}{s-a}$; $r_2 = \frac{\Delta}{s-b}$; $r_3 = \frac{\Delta}{s-c}$;

(ii) $r_1 = s\tan\frac{A}{2}$; $r_2 = s\tan\frac{B}{2}$; $r_3 = s\tan\frac{C}{2}$

(iii) $r_1 = \frac{a\cos\frac{B}{2}\cos\frac{C}{2}}{\cos\frac{A}{2}}$ and so on **(iv)** $r_1 = 4R\sin\frac{A}{2}.\cos\frac{B}{2}.\cos\frac{C}{2}$ and so on

Illustration 18

In a triangle ABC, find the value of $\frac{b-c}{r_1}+\frac{c-a}{r_2}+\frac{a-b}{r_3}$.

Solution:

$$\frac{b-c}{\left(\frac{\Delta}{s-a}\right)}+\frac{c-a}{\left(\frac{\Delta}{s-b}\right)}+\frac{a-b}{\left(\frac{\Delta}{s-c}\right)}$$

$= \frac{1}{\Delta}[(b-c)(s-a)+(c-a)(s-b)+(a-b)(s-c)]$

$= \frac{1}{\Delta}[s(b-c+c-a+a-b)-a(b-c)-b(c-a)-c(a-b)] = 0$

Illustration 19

In an acute angled triangle ABC, $r + r_1 = r_2 + r_3$ and $\angle B > \frac{\pi}{3}$, then prove that $b + 3c < 3a < 3b + 3c$

Solution:

$r - r_2 = r_3 - r_1 \Rightarrow \frac{\Delta}{s}-\frac{\Delta}{s-b}=\frac{\Delta}{s-c}-\frac{\Delta}{s-a}$

or $\frac{-b}{s(s-b)}=\frac{c-a}{(s-c)(s-a)}$

or $\frac{(s-a)(s-c)}{s(s-b)} = \frac{a-c}{b} \Rightarrow \tan^2\frac{B}{2} = \frac{a-c}{b}$

But $\frac{B}{2} \in \left(\frac{\pi}{6},\frac{\pi}{4}\right)$. Therefore,

$\tan^2\frac{B}{2} \in \left(\frac{1}{3},1\right) \Rightarrow \frac{1}{3} < \frac{a-c}{b} < 1$

or $b < 3a - 3c < 3b$

$b + 3c < 3a < 3b + 3c$

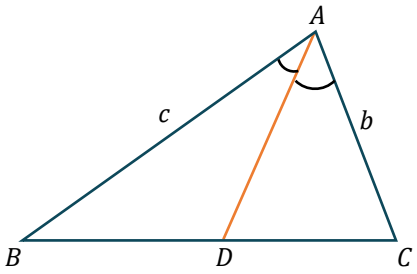
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10. ANGLE BISECTORS & MEDIANS AND ALTITUDE:

(i) An angle bisector divides the base in the ratio of corresponding sides.

$$\frac{BD}{CD}=\frac{c}{b} \Rightarrow BD=\frac{ac}{b+c} \ \& \ CD=\frac{ab}{b+c}$$



(ii) If m_a and β_a are the lengths of a median and an angle bisector from the angle A then,

$$m_a=\frac{1}{2}\sqrt{2b^2+2c^2-a^2} \text{ and } \beta_a=\frac{2bc\cos\frac{A}{2}}{b+c}$$

Note that $m_a^2+m_b^2+m_c^2=\frac{3}{4}(a^2+b^2+c^2)$

(iii) Length of altitude from the angle $A = A_a = \frac{2\Delta}{a}$

Illustration 20

In ΔABC, AD & BE are its two median. If $AD = 4$, $\angle DAB = \frac{\pi}{6}$ and $\angle ABE = \frac{\pi}{3}$ then find the length of BE and area of ΔABC.

Solution:

$AP = \frac{2}{3}$; $AD = \frac{8}{3}$; $PD = \frac{4}{3}$; Let $PB = x$

$\tan 60° = \frac{8/3}{x}$ or $x=\frac{8}{3\sqrt{3}}$

Area of $\Delta ABP = \frac{1}{2} \times \frac{8}{3} \times \frac{8}{3\sqrt{3}} = \frac{32}{9\sqrt{3}}$

$\therefore$ Area of $\Delta ABC = 3 \times \frac{32}{9\sqrt{3}} = \frac{32}{3\sqrt{3}}$

Also, $BE = \frac{3}{2}x = \frac{4}{\sqrt{3}}$

Illustration 21

If AD, BE and CF are the medians of a ΔABC, then $(AD^2 + BE^2 + CF^2) : (BC^2 + CA^2 + AB^2)$ is equal to

(A) 4 : 3 (B) 3 : 2 (C) 3 : 4 (D) 2 : 3

Ans. (C)

Solution:

$\because AD^2 = \frac{1}{4}(2b^2 + 2c^2 - a^2)$,

$BE^2 = \frac{1}{4}(2c^2 + 2a^2 - b^2)$ and

$CF^2 = \frac{1}{4}(2a^2 + 2b^2 - c^2)$ $\therefore \frac{AD^2+BE^2+CF^2}{BC^2+CA^2+AB^2} = \frac{3}{4}$

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Illustration 22

If p_1, p_2, p_3 are respectively the lengths of perpendiculars from the vertices of a triangle ABC to the opposite sides, prove that :

(i) $\frac{\cos A}{p_1}+\frac{\cos B}{p_2}+\frac{\cos C}{p_3}=\frac{1}{R}$ (ii) $\frac{bp_1}{c}+\frac{cp_2}{a}+\frac{ap_3}{b}=\frac{a^2+b^2+c^2}{2R}$

Solution:

(i) use $\frac{1}{p_1}=\frac{a}{2\Delta},\frac{1}{p_2}=\frac{b}{2\Delta},\frac{1}{p_3}=\frac{c}{2\Delta}$

$\therefore$ LHS = $\frac{1}{2\Delta}$ $(a\cos A + b\cos B + c\cos C)$

$= \frac{R}{2\Delta}(\sin 2A + \sin 2B + \sin 2C) = \frac{4R\sin A\ \sin B\ \sin C}{2\Delta}$

$= \frac{4R}{2\Delta}\cdot\frac{a}{2R}.\frac{b}{2R}.\frac{c}{2R}=\frac{1}{4\Delta R^2}abc=\frac{1}{4\Delta R^2}.(4R\Delta)=\frac{1}{R}$ = RHS

(ii) LHS = $\frac{bp_1}{c}+\frac{cp_2}{a}+\frac{ap_3}{b}=\frac{a^2+b^2+c^2}{2R}=\frac{2b\Delta}{ac}+\frac{2c\Delta}{ab}+\frac{2a\Delta}{bc}=\frac{2\Delta(a^2+b^2+c^2)}{abc}$

$= \frac{2\Delta(a^2+b^2+c^2)}{4\Delta R}=\frac{a^2+b^2+c^2}{2R}$

11. AMBIGUOUS CASE:

When two sides and an angle opposite to one of them is given (Ambiguous case):

By applying cosine rule, we have $\cos B = \frac{a^2+c^2-b^2}{2ac}$

$\Rightarrow a^2 - (2c\cos B)a + (c^2 - b^2) = 0 \Rightarrow a = c\cos B \pm\sqrt{(c\cos B)^2-(c^2-b^2)}$

$\Rightarrow a = c\cos B \pm \sqrt{b^2-(c\sin B)^2}$

This equation leads to following cases :

Case-I : If $b < c\sin B$, no such triangle is possible.

Case-II : Let $b = c\sin B$. There are further following case :

(a) B is an obtuse angle $\Rightarrow \cos B$ is negative. There exists no such triangle.

(b) B is an acute angle $\Rightarrow \cos B$ is positive. There exists only one such triangle.

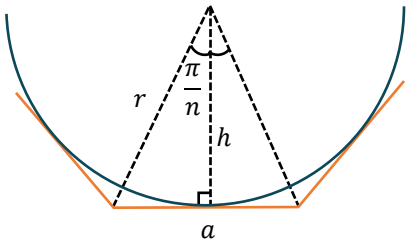
Case-III : Let $b > c\sin B$. There are further following cases :

(a) B is an acute angle $\Rightarrow \cos B$ is positive. In this case triangle will exists if and only if $c\cos B\sqrt{b^2-(c\sin B)^2}$ > or $c > b \Rightarrow$ Two such triangle is possible. If $c < b$, only one such triangle is possible.

(b) B is an obtuse angle $\Rightarrow \cos B$ is negative. In this case two values of a will exist if and only if $\sqrt{b^2-(c\sin B)^2} > |c\cos B| \Rightarrow b > c$. So in this case only one such triangle is possible. If $b < c$ there exists no such triangle.

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12. REGULAR POLYGON:



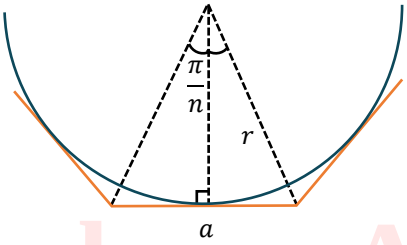
A regular polygon has all its sides equal. It may be inscribed or circumscribed.

(a) Inscribed in Circle of Radius r:

(i) $a = 2h \ \tan\frac{\pi}{n} = 2r\sin\frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides inscribed in a circle of radius r are given by $P = 2nr\sin\frac{\pi}{n}$ and $A = \frac{1}{2}nr^2\sin\frac{2\pi}{n}$

(b) Circumscribed About a Circle of Radius r:



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(i) $a = 2r\tan\frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides circumscribed about a given circle of radius r is given by $p = 2nr\tan\frac{\pi}{n}$ and $A = nr^2\tan\frac{\pi}{n}$

Illustration 23

A regular hexagon & a regular dodecagon are inscribed in the same circle. If the side of the dodecagon is $\left(\sqrt{3}-1\right)$, if the side of the hexagon is $\sqrt[4]{k}$, then find value of k.

Ans. $\sqrt{2}$

Solution:

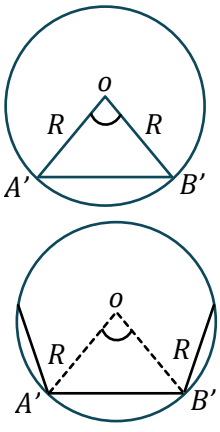
For dodecagon $\angle A'OB' = \frac{2\pi}{12} = 30°$

$\Rightarrow \angle OA'B' = \angle OB'A' = 75° \Rightarrow \frac{R}{\sin 75°} = \frac{\sqrt{3}-1}{\sin 30°}$

$\Rightarrow R = \frac{(\sqrt{3}-1)\ (\sqrt{3}+1)}{2\sqrt{2}\times\frac{1}{2}} \Rightarrow R = \sqrt{2}$

For hexagon $\angle AOB = \frac{2\pi}{6} = 60°$

$\Rightarrow \Delta AOB$ is equilateral $\Rightarrow AB = R = \sqrt{2}$



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